

MIHIR MOHITE

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PROFESSIONAL SUMMARY

B.Tech AI/ML student at MIT-WPU (CGPA 8.62) with **two technical internships** and a pilot-stage construction-tech product (ForeSites), covering **RAG retrieval, agentic LLM workflows, and serverless backends on AWS**. Working languages: **C++** and **Python**.

WORK EXPERIENCE

Machine Learning Intern | BeyondBot Technology Pvt. Ltd. Oct 2025 – Present | On-site Pune

- Delivered a legal-compliance **RAG chatbot** PoC over a **1,210-page** corpus of Indian industrial, manufacturing, and environmental law for on-site customer queries; PoC transitioned to platform engineers for product integration.
- Architected the pipeline end to end: **Mistral OCR** → custom **regex router** (definitions, sections, tables, amendments) → **LightRAG** (KG + vector) with **SQLite** tabular lookups and a JSON amendment index → **Agno agent** (Gemini 3 Flash, 4 tools) → **FastAPI** with SSE streaming → chat UI with PDF.js viewer and citation badges.
- 49% retrieval-quality improvement (0.77 vs. 0.51 RAGAS, hybrid mode over naive baseline)** demonstrated via a RAGAS-style evaluation harness benchmarking **5 LightRAG retrieval modes** across **315 queries** on a 10k-excerpt FinDER subset — measuring faithfulness, answer relevancy, context precision/recall, retrieval latency, token count, and deployment cost.

Technologies / Skills Used : Python, FastAPI, LightRAG, SQLite, Mistral OCR, Agno agent

Software Intern | Heera Software Private Limited Jul 2025 – Oct 2025 | On-site Pune

- Scored address confidence for **~13,500 records** by developing a validation service on the **Google Maps API** with **fuzzy matching** and **reverse-geocoded cosine similarity**.
- Improved logistics trust for clients including **Red Bull** by providing a rapid, repeatable address-confidence scoring method.

Technologies / Skills Used : Google Maps API, Python, Machine Learning, Statistical Analysis

EDUCATION

Year	Degree	Institute	CGPA/MARKS
Jul 2023 – May 2027	B.Tech in Electronics & Communication Engineering, Specialization in AI & ML	MIT-WPU	8.62/10.00

PROJECTS

ForeSites | Team Member

- Built streaming **NL-to-SQL** over a **9-table Postgres** schema via a Python **AWS Lambda** agent (**Gemini API**), with per-session memory and session-ownership enforcement.
- Built the **WhatsApp ingestion service** (**Node.js/Express** + **AWS S3**) for snag reporting via text, voice, and image, backed by **Supabase**.
- Redesigned the **React** dashboard frontend; authored the relational schema and snag-assignment workflow.

Technologies / Skills Used : Python, AWS Lambda, Node.js, Express, PostgreSQL, React

AgroSense | Team Member

- Processed UAV RGB imagery into orthomosaics with **ExG/VARI vegetation indices** and **K-Means zoning** via a **photogrammetry** pipeline; integrated an **ESP32** ground module with a Modbus soil sensor, GPS waypoint guidance, and on-device ML for **NPK estimation**, reporting to a **Firebase/Vercel** dashboard with a multilingual AI assistant.
- Authored a review paper on affordable, industry-standard **precision agriculture** adoption for smallholder farms in Maharashtra.

Technologies / Skills Used : ESP32, Firebase, Vercel, React, Node.js, MongoDB

EKG | Solo Project

- Developed a mini-graph PoC in **Neo4j** ingesting SQL, JSON, and TXT data behind a **Gemini NL-query** interface, letting enterprises query disconnected databases for interconnected context in natural language.

Technologies / Skills Used : Neo4j, React, Node.js, MongoDB

SKILLS

Programming Languages : C++, Python, JavaScript, SQL
Frameworks & Libraries : FastAPI, Node.js, React, TensorFlow, Keras, PyTorch, OpenCV
Databases : PostgreSQL, MySQL
Tools & Platforms : AWS, GCP, Google Maps API, SQLite, Supabase, Neo4j, Git, MATLAB
Core CS : Data Structures, Algorithms, Complexity Analysis, Object-Oriented Design

LEADERSHIP & EXTRACURRICULARS

President, CoDeC — Competitive Programming Club | MIT-WPU Oct 2025 – Present | Pune

- 350+ participants (87 teams)** from MIT-WPU and external engineering colleges hosted at Trifecta Challenge 2026, a 3-day flagship symposium; led a core team of **15–20**, owning **track design, scheduling, judge coordination, and day-of operations** across Full-Stack, ML, and Competitive Programming tracks.
- Partnerships with **GeeksforGeeks, HackerRank, and AlgoZenith** secured through direct **outreach and negotiation**; managed sponsor deliverables and external-college registrations end to end.
- Set the club’s annual **roadmap and technical vision**; run recurring workshops and developer events on **DSA** and **competitive programming**.